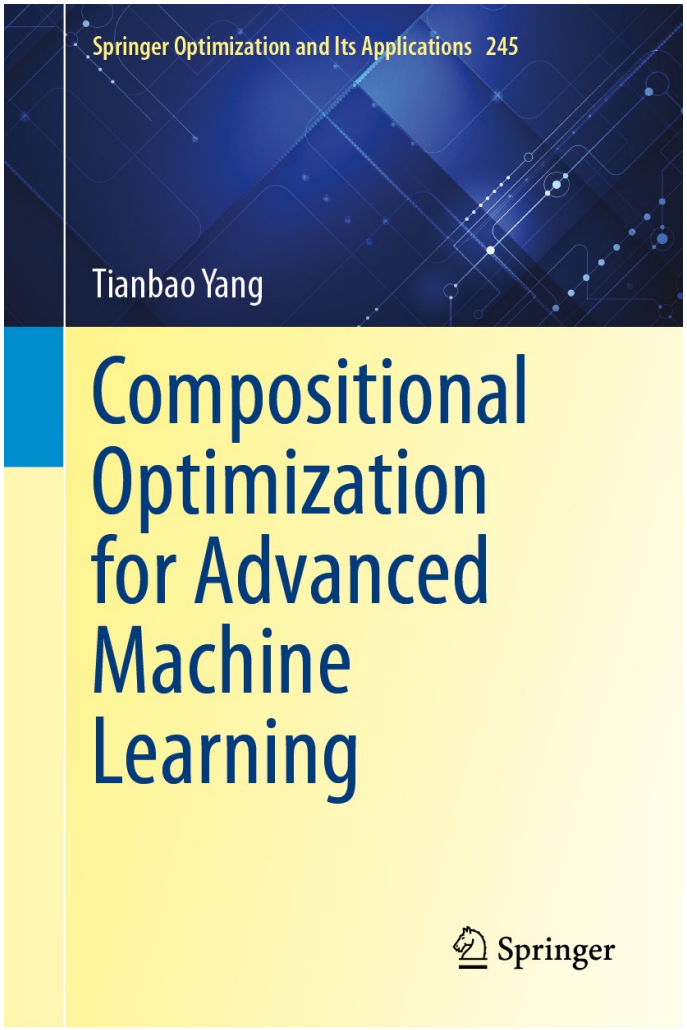
Compositional Optimization for Advanced Machine Learning

Tianbao Yang
Professor, Stephen Horn '79 Engineering Excellence Chair
CSE@TAMU

Roadmap

- Objectives & Frameworks
- Algorithms & Theories
- Applications & Software

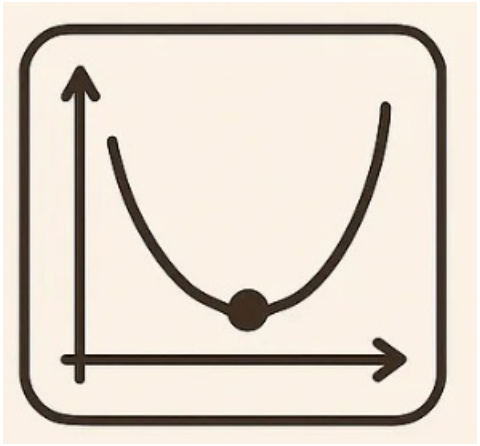


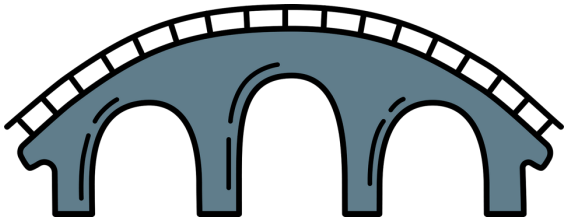
My research philosophy

“The best theory is inspired by practice. The best practice is inspired by theory” - Donald Knuth

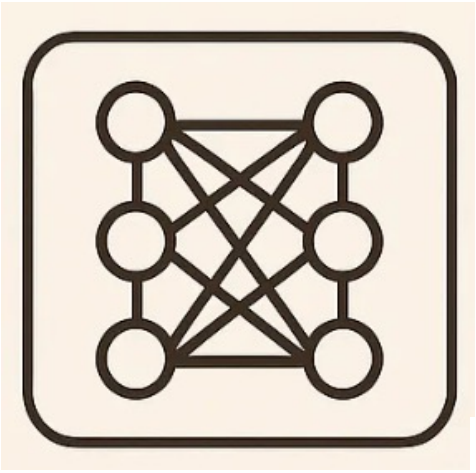
“知者行之始，行者知之成”
- 王阳明

Optimization

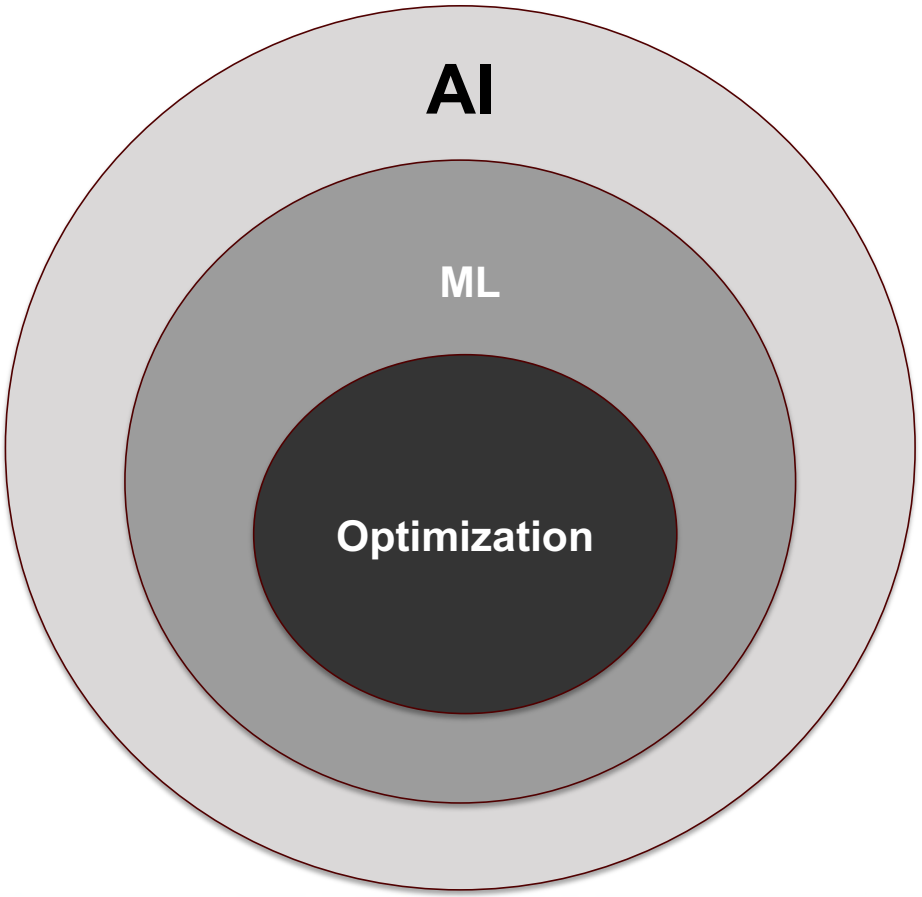




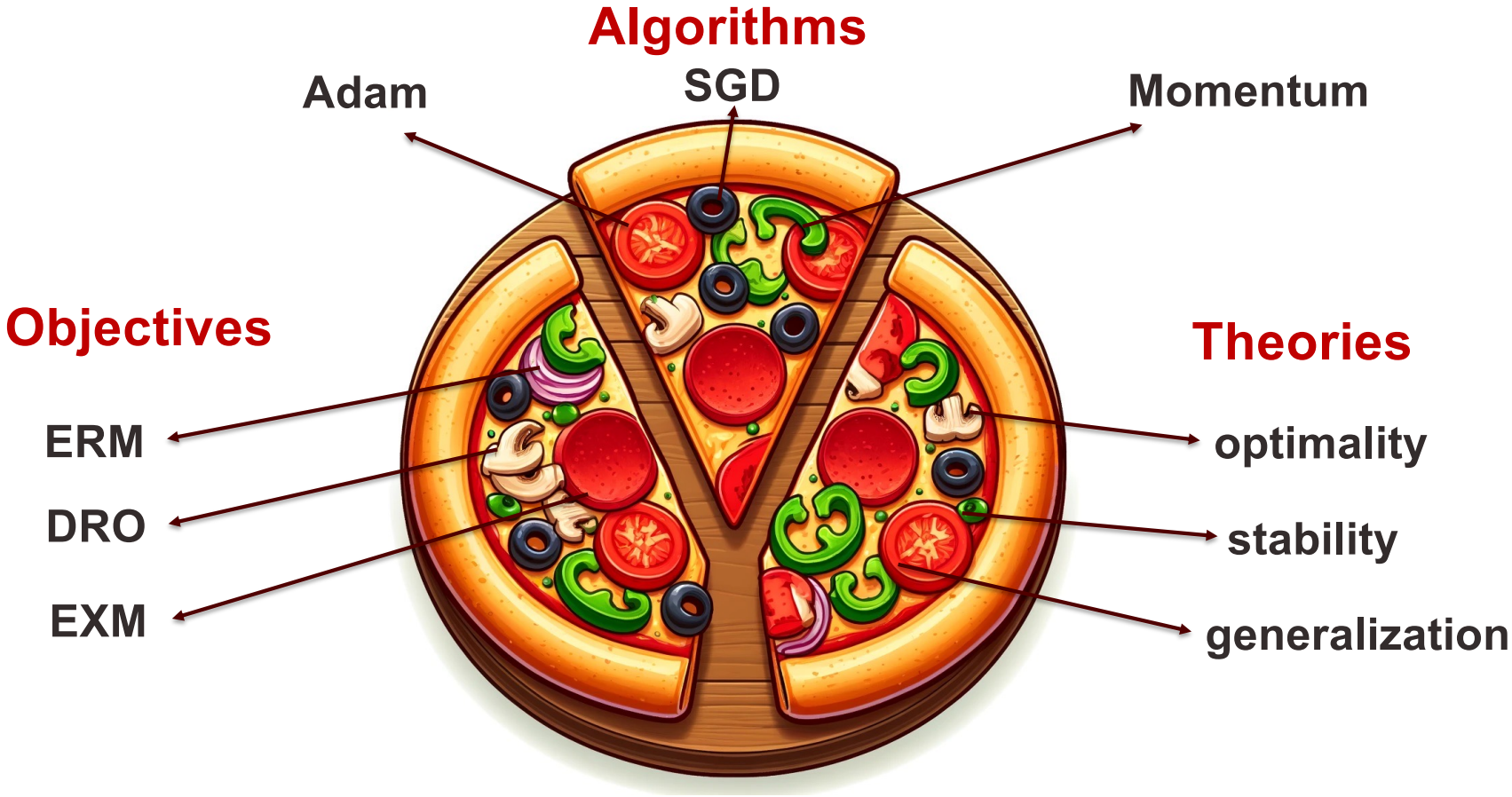
ML/AI



AI, ML and Optimization



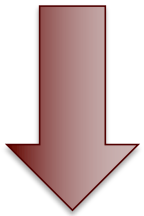
Optimization for ML/AI



Objectives

Empirical Risk Minimization (ERM)

$$\min_{\mathbf{w}} F(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^{n} \ell(\mathbf{w}, \mathbf{x}_i, y_i)$$



$$\min_{\mathbf{w}} \mathbb{E}_{\xi}[f(\mathbf{w}, \xi)]$$ **Stochastic Optimization**

Train the model with Mini-batching

STANDARD PYTORCH PIPELINE

MINI-BATCH LOSS

$$\frac{1}{B}\sum_{i\in\mathcal{B}}\ell(\mathbf{w},\mathbf{x}_i,y_i)$$

You Define the mini-batch loss.

1 Compute mini-batch loss

```
loss_fn = CrossEntropyLoss()
loss = loss_fn(preds, target)
```

2 Backpropagate gradients

```
loss.backward()
```

3 Update model parameters

```
optimizer.step()
```

PyTorch handles the rest.

Cross-Entropy loss for Classification

$$\ell(\mathbf{w}, \mathbf{x}, y) = -\log \frac{\exp([h(\mathbf{w}; \mathbf{x})]_y)}{\sum_{l=1}^{K} \exp([h(\mathbf{w}; \mathbf{x})]_l)}$$

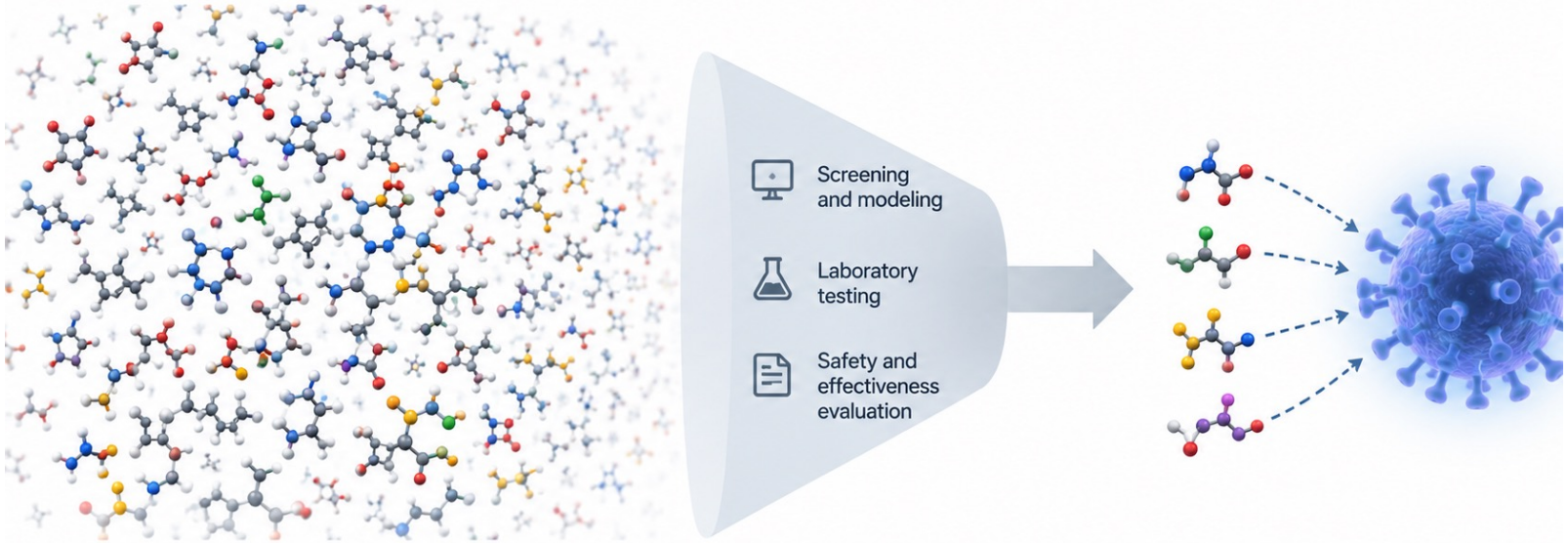
$\mathbf{x}$: input data, $\quad y \in \{1, \ldots, K\}$: label,

$h(\mathbf{w}; \mathbf{x}) \in \mathbb{R}^K$: prediction scores

What if K is million even billion?

Drug discovery

To design a drug against a new virus, we need to identify the right molecules from millions of them.



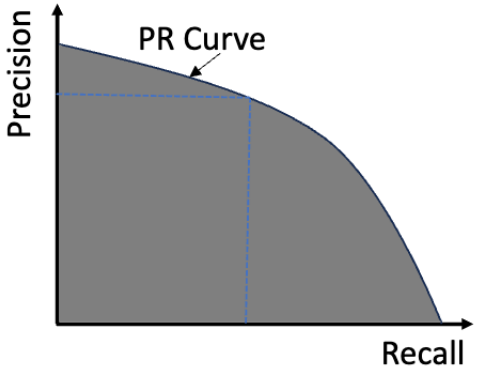
Millions of molecules
A vast library of potential drug candidates

Only a few molecules
have the right properties to fight the virus

MIT AICures Challenge

Evaluation Metric: AUPRC

Fighting Secondary Effects of Covid



(a) Test PRC-AUC

Rank	Model	Author	Submissions	Test PRC-AUC
1	MolecularG	AIDrug@PA	7	0.725
2	-	AGL Team	20	0.702
3	MoleculeKit	DIVE@TAMU	7	0.677
4	GB	BI	6	0.67
5	Chemprop ++	AICures@MIT	4	0.662
6	-	Mingjun Liu	3	0.657
7	Pre-trained OGB-GIN (ensemble)	Weihua Hu@Stanford	2	0.651
8	RF + fingerprint	Cyrus Maher@Vir Bio	1	0.649
9	Graph Self-supervised Learning	SJTU_NRC_Mila	3	0.622
10	-	Congjie He	10	0.611

(b) Test ROC-AUC

Rank	Model	Author	Submissions	Test ROC-AUC
1	MoleculeKit	DIVE@TAMU	7	0.928
2	Chemprop ++	AICures@MIT	4	0.877
3	-	Gianluca Bontempi	7	0.848
4	-	Apoorv Umang	1	0.84
5	Pre-trained OGB-GIN (ensemble)	Weihua Hu@Stanford	2	0.837
6	-	Kexin Huang	1	0.824
7	Chemprop	Rajat Gupta	7	0.818
8	MLP	IITM	7	0.807
9	Graph Self-supervised Learning	SJTU_NRC_Mila	3	0.8
10	-	Congjie He	10	0.8

Empirical Estimator: Average Precision (AP)

$$\mathrm{AP}(h) = \frac{1}{n_+} \sum_{\mathbf{x}_i \in \mathcal{S}_+} \mathrm{Precision}(h(\mathbf{x}_i))$$

Empirical Estimator: Average Precision (AP)

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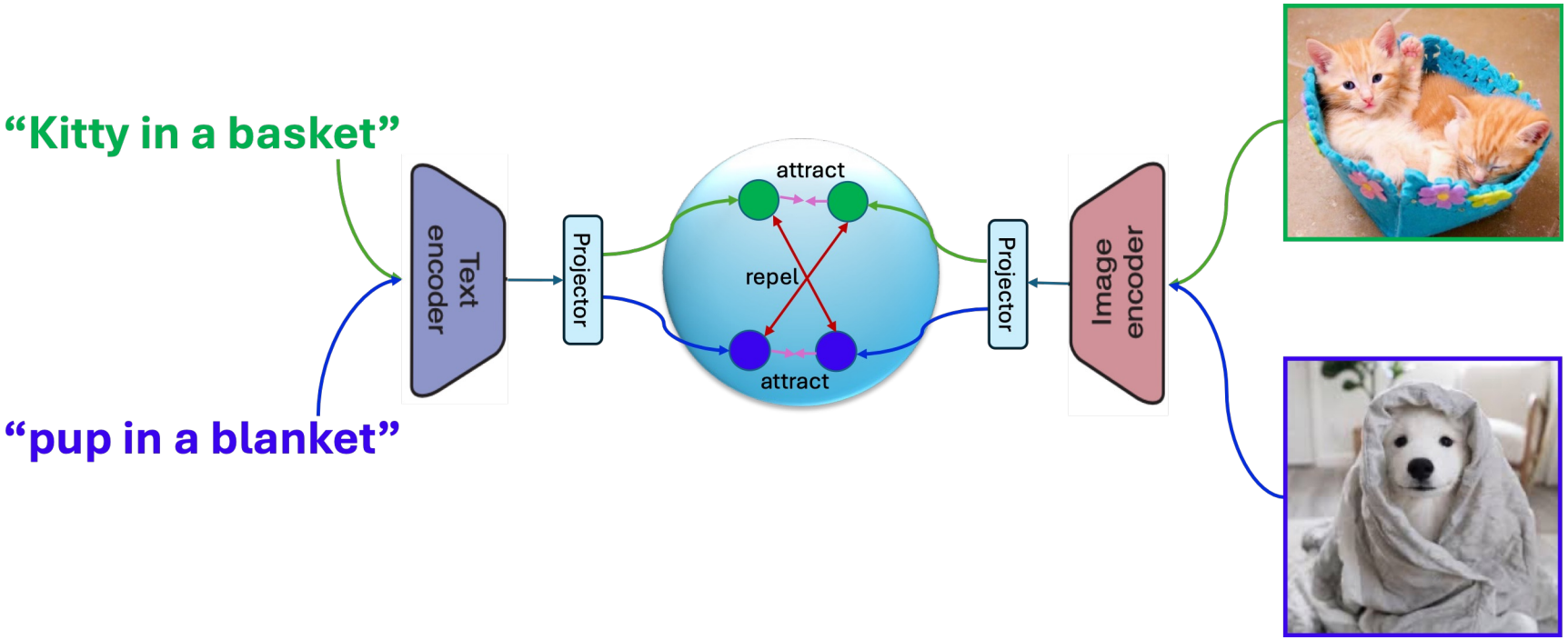
Positive Examples

$$\mathrm{Precision}(h(\mathbf{x}_i)) = \frac{\sum_{\mathbf{x}_j \in \mathcal{S}_+} \mathbb{I}(h(\mathbf{x}_j) \geq h(\mathbf{x}_i))}{\sum_{j=1}^{n} \mathbb{I}(h(\mathbf{x}_j) \geq h(\mathbf{x}_i))}$$

All Examples

What if n is million even billion?

Foundation Model by Self-supervised Learning



OpenAI’s CLIP (Radford et al. ’21)

Mini-batch Contrastive Loss

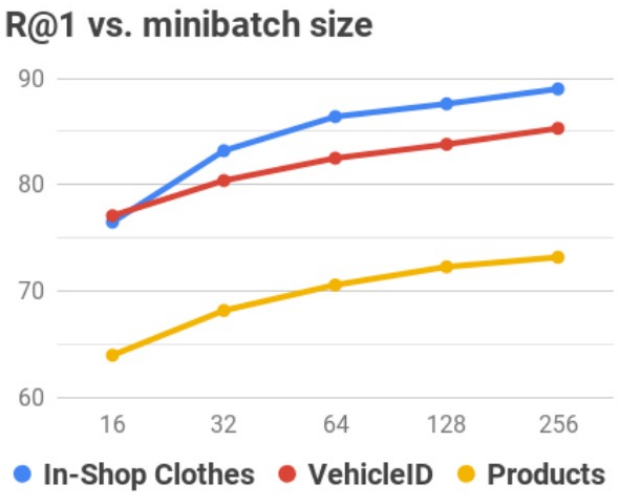
$$\ell(\mathbf{w}, \mathbf{x}_i) = -\tau \log \left(\frac{\exp(h(\mathbf{w}; \mathbf{x}_i)^\top h(\mathbf{w}; \mathbf{t}_i)/\tau)}{\sum_{j \in \mathcal{B}_i^-} \exp(h(\mathbf{w}; \mathbf{x}_i)^\top h(\mathbf{w}; \mathbf{t}_j)/\tau)} \right)$$

Mini-batch
Negative data

$h(\mathbf{w}; \mathbf{x}) \in \mathbb{R}^d$: embedding vector

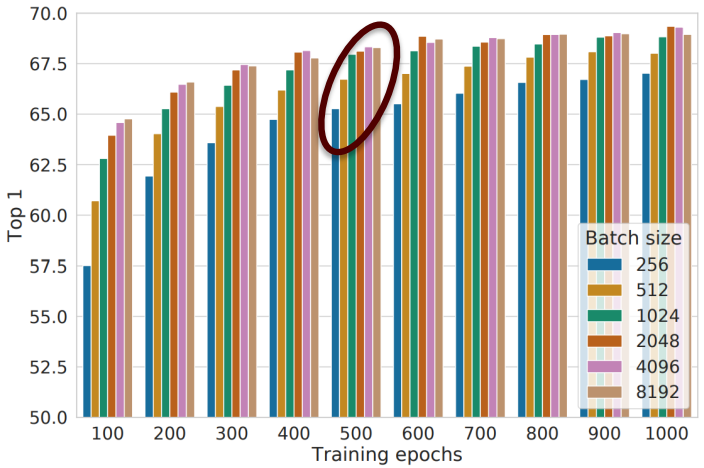
The Issue: Large batch size

Optimizing ranking measures



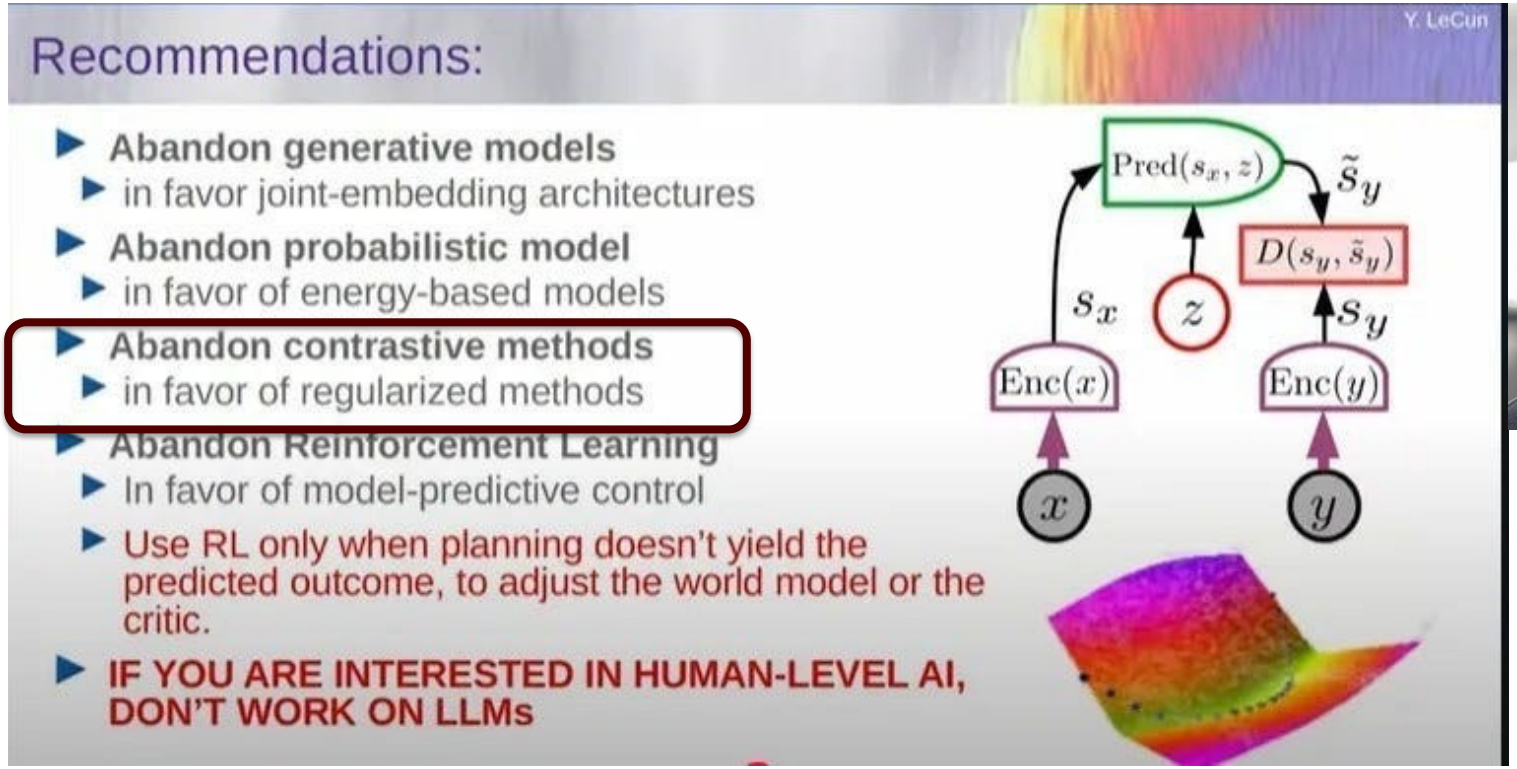
FastAP: Cakir et al. 2019

Self-supervised Learning



SimCLR: Chen et al. 2020.

To Contrast or Not to Contrast?

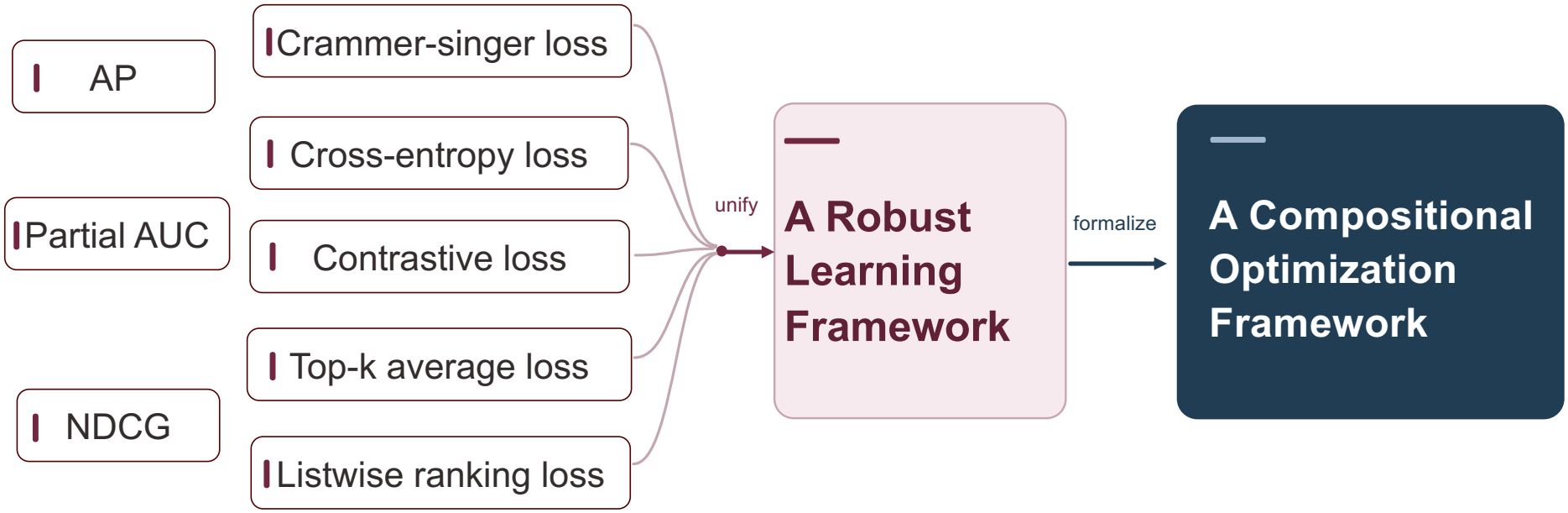




Yann Lecun

Frameworks

We need new frameworks



A Unified Robust Learning Framework

classification **ranking** **repre. learning**

$$s(\mathbf{w}; \mathbf{x}, \mathbf{y})$$

Anchor space X



Anchor x

"a golden retriever dog"

Contrasting-object space Y

The contrasting object y can be a label, image, text

	Few positives y+		**Many negatives y–**				
Label objects	golden retriever (ID:1)	...	Cat (ID:2)	Bird (ID:3)	Horse (ID:4)	Airplane (ID:5)	...
Image objects	 	...	    	...			
Text objects	*"a golden retriever dog"*	...	*"a tabby cat"*	*"a blue bird"*	*"a horse"*		...

Aggregating over negatives

Pairwise Loss

$$\ell_{ij}(\mathbf{w}) = s(\mathbf{w}; \mathbf{x}_i, \mathbf{y}_j^-) - s(\mathbf{w}; \mathbf{x}_i, \mathbf{y}_i^+)$$

Average of Negatives

$$\frac{1}{K} \sum_{j=1}^{K} \ell_{ij}(\mathbf{w})$$

Treat all negatives equally

All Negatives

Distributionally Robust Optimization (DRO)

$$L_i(\mathbf{w}) = \max_{\mathbf{p} \in \Delta} \sum_{j=1}^{K} p_j \ell_{ij}(\mathbf{w}) - \tau \mathrm{KL}(\mathbf{p}, \frac{1}{K})$$

Probability weights

Hard negatives get higher weights

$$L_i(\mathbf{w}) = \tau \log \left(\frac{1}{K} \sum_{j=1}^{K} \exp(\ell_{ij}(\mathbf{w})/\tau) \right)$$

A Unified Robust Learning Framework

$$L_i(\mathbf{w}) = \max_{\mathbf{p}\in\Delta} \sum_{j=1}^{K} p_j \ell_{ij}(\mathbf{w}) - \tau \mathrm{KL}(\mathbf{p}, \frac{1}{K})$$

Cross-entropy loss



dog

$$s(\mathbf{w}; \mathbf{x}, y) = \phi(\mathbf{w}; \mathbf{x})^\top \mathbf{w}_y$$

Contrastive loss



"this is a dog"

$$s(\mathbf{w}; \mathbf{x}, \mathbf{y}) = h(\mathbf{w}_1; \mathbf{x})^\top h(\mathbf{w}_2; \mathbf{y})$$

Listwise CE loss

"dog"









1. Vary scoring function

2. Vary or Optimize Temperature

3. Vary divergence measure

The Optimization Challenge

$$\min_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^{n} \tau \log \left(\frac{1}{K} \sum_{j=1}^{K} \exp(\ell_{ij}(\mathbf{w})/\tau) \right)$$

Large sum

A Compositional Optimization Framework

X-risk

A family of **Compositional Risks** in which the loss function of each data point is defined in a way that **contrasts** the data point with **many** others.

$$\min_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^{n} f_i(g(\mathbf{w}, \mathbf{x}_i, \mathcal{S}_i))$$

$$g(\mathbf{w}, \mathbf{x}_i, \mathcal{S}_i) = \frac{1}{|\mathcal{S}_i|} \sum_{\mathbf{y} \in \mathcal{S}_i} \ell(\mathbf{w}; \mathbf{x}_i, \mathbf{y})$$

Empirical X-risk Minimization

Yuan et al. (KDD'23)

Map Contrastive Objectives to X-risk

$$\frac{1}{n}\sum_{i=1}^{n}\tau\log\left(\frac{1}{K}\sum_{j=1}^{K}\exp(\ell_{ij}(\mathbf{w})/\tau)\right)$$

$$\min_{\mathbf{w}}\frac{1}{n}\sum_{i=1}^{n}f_i(g(\mathbf{w},\mathbf{x}_i,\mathcal{S}_i))$$

Finite-sum Coupled Compositional Optimization (FCCO)

$$\min_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^{n} f_i\left(\mathbb{E}_{\zeta \sim \mathbb{P}_i}\left[g_i(\mathbf{w}; \zeta)\right]\right)$$

Coupling

Wang & Yang. (ICML'22)

Algorithms & Theories

Biased Stochastic Gradient

Not Practical

$$\nabla f_i(g(\mathbf{w}, \mathbf{x}_i, \mathcal{S}_i))\nabla g(\mathbf{w}, \mathbf{x}_i, \mathcal{S}_i)$$

$$\mathbb{E} \neq$$

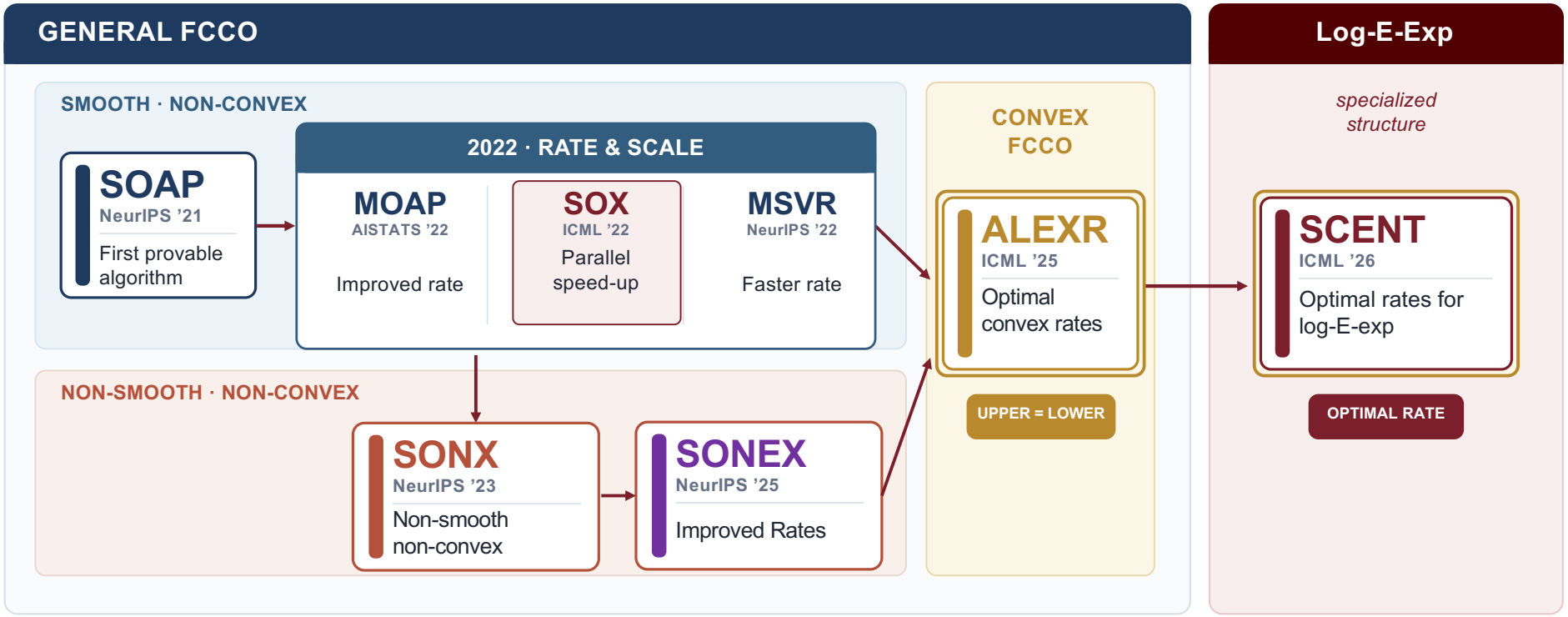
Inner Mini-batch

$$\nabla f_i(g(\mathbf{w}, \mathbf{x}_i, \mathcal{B}_i))\nabla g(\mathbf{w}, \mathbf{x}_i, \mathcal{B}_i)$$

Large Batch Size

(Hu et al.'19)

A Complete Algorithmic & Theory Landscape



Seven algorithms, one unified research program

SOX

Gradient **E**stimator

$$\mathbf{z}_t = \frac{1}{B_1} \sum_{\mathbf{x}_i \in \mathcal{B}_1^t} \nabla f_i(u_{i,t}) \nabla g(\mathbf{w}_t, \mathbf{x}_i, \mathcal{B}_i)$$

Estimator of g_i

$$u_{i,t} = \begin{cases} (1-\gamma_t) u_{i,t-1} + \gamma_t g(\mathbf{w}_t, \mathbf{x}_i, \mathcal{B}_i) & \mathbf{x}_i \in \mathcal{B}_1^t \\ u_{i,t-1} & \text{o.w.} \end{cases}$$

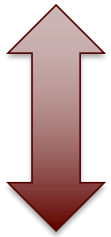
Moving-**A**verage

$$\mathbf{v}_t = (1-\beta_t)\mathbf{v}_{t-1} + \beta_t \mathbf{z}_t$$

Wang & Yang. (ICML'22)

SCENT

$$\min_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^{n} \log \left(\mathbb{E}_{\zeta \sim \mathbb{P}_i} \exp(s_i(\mathbf{w}; \zeta)) \right)$$



$$\min_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^{n} \min_{\nu_i} \left\{ e^{-\nu_i} \left(\mathbb{E}_{\zeta \sim \mathbb{P}_i} \exp(s_i(\mathbf{w}; \zeta)) \right) + \nu_i \right\}$$

Wei et al. (ICML'26)

SCENT

stochastic proximal mirror descent (SPMD) $\varphi(\nu) = e^{-\nu}$

$$\nu_{i,t} = \begin{cases} \arg\min_\nu e^{-\nu_i} e^{s_i(\mathbf{w}_t;\zeta_t)} + \frac{D_\varphi(\nu,\nu_{i,t-1})}{\alpha_t} & i \in \mathcal{B}_1^t \\ \nu_{i,t-1} & i \notin \mathcal{B}_1^t \end{cases}$$

Recovers SOX

$$u_{i,t} = e^{\nu_{i,t}}, \alpha_{i,t} = e^{-\nu_{i,t-1}}$$

Complexity Theory of SOX

Assumption: Non-convex & Smooth

Iteration Complexity

$$\mathbb{E}\|\nabla F(\mathbf{w})\|^2 \leq \epsilon^2 \quad \Rightarrow \quad O\left(\frac{n}{B_1\epsilon^4}\right)$$

Better **T**otal **C**omplexity than BSGD when

$$n \leq \frac{1}{\epsilon^2}$$

Wang & Yang. (ICML'22)

Complexity Theory of SCENT

Assumption: Convex & Non-smooth

Iteration Complexity

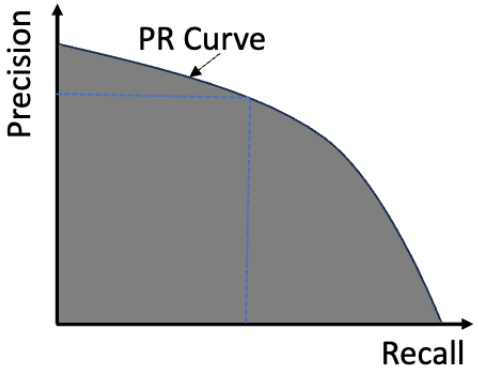
$$\mathbb{E}[F(\mathbf{w}) - F_*] \leq \epsilon \quad \Rightarrow \quad O\left(\frac{n}{B_1 \epsilon^2}\right)$$

Wei et al. (ICML'26)

Applications & Software

1st Place at MIT AICures Challenge

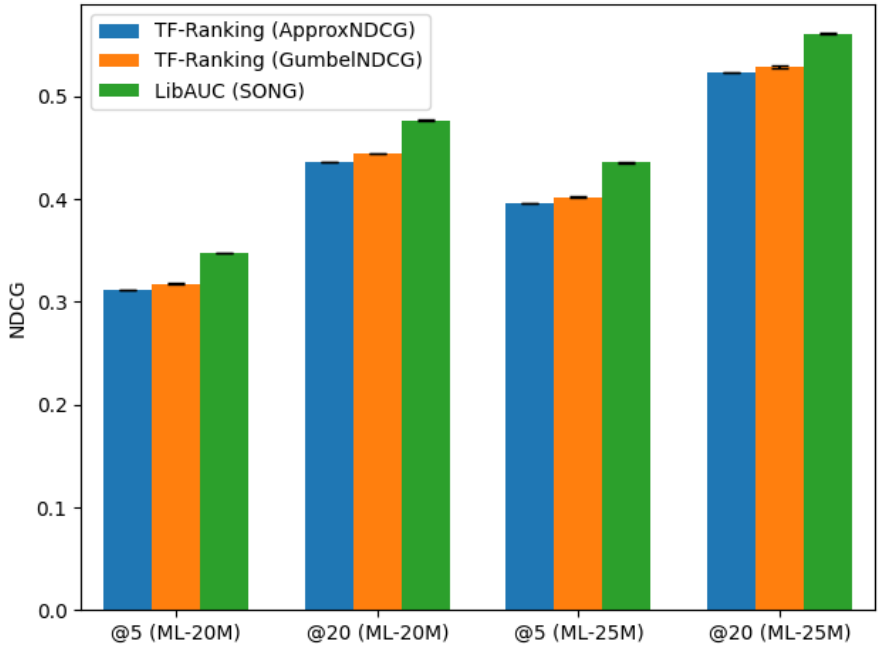
Fighting Secondary Effects of Covid

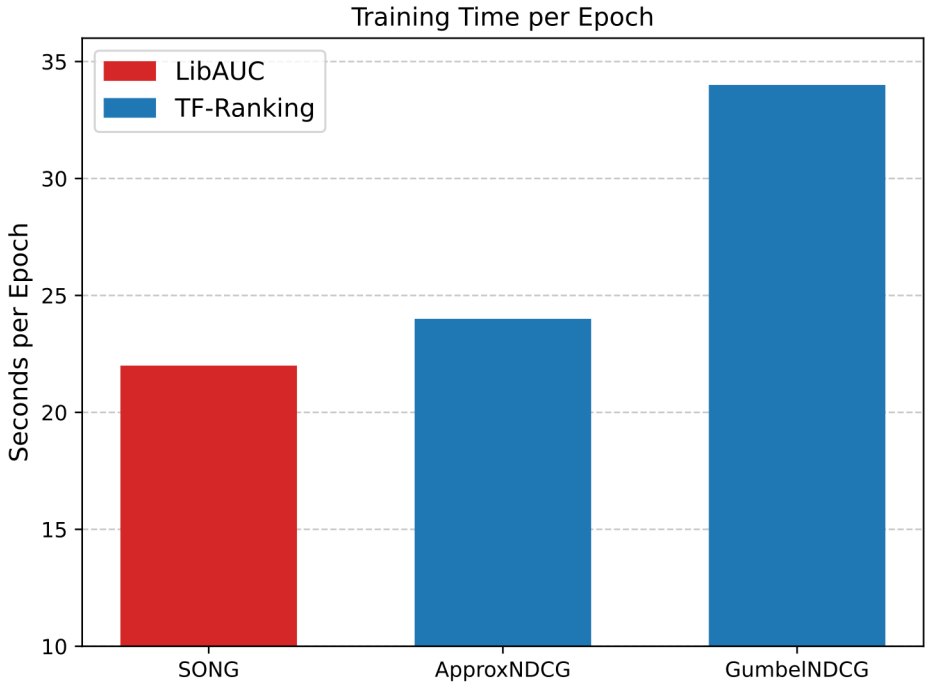


Rank	Model	Author	Submissions	10-fold CV ROC-AUC	10-fold CV PRC-AUC	Test ROC-AUC	Test PRC-AUC
1		DIVE@TAMU	11			0.957	0.729
2	MolecularG	AIDrug@PA	9			0.7	0.725
3		AGL Team	20			0.675	0.702
4		phucdoitoan@Fujitsu	14	0.898 +/- 0.113	0.508 +/- 0.253	0.867	0.694
5	GB	BI	6			0.698	0.67
6	Chemprop ++	AICures@MIT	4			0.877	0.662
7		Mingjun Liu	3			0.72	0.657
8	Pre-trained OGB-GIN (ensemble)	Weihua Hu@Stanford	2	0.905 +/- 0.133	0.494 +/- 0.333	0.837	0.651
9	RF + fingerprint	Cyrus Maher@Vir Bio	1	0.896 +/- 0.074	0.481 +/- 0.338	0.799	0.649
10	Graph Self-supervised Learning	SJTU_NRC_Mila	3	0.825 +/- 0.210	0.530 +/- 0.342	0.800	0.622

Wang et al. (Bioinformatics'22)

NDCG Optimization

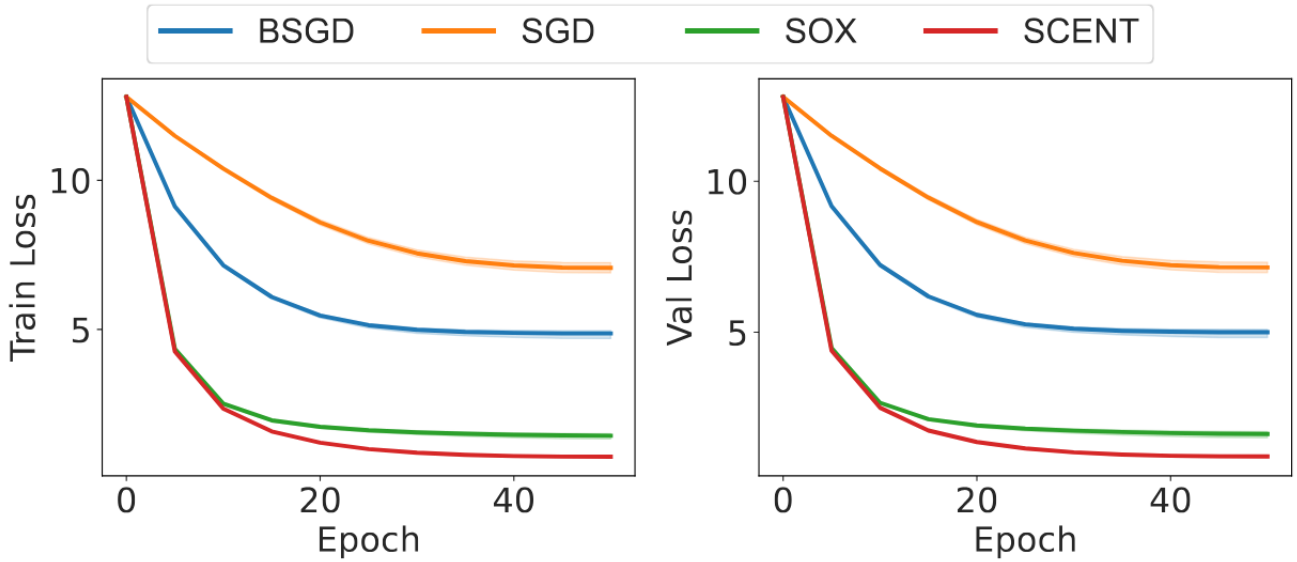




MovieLens (ML) 20M and 25M datasets

Yuan et al. (KDD'23)

Extreme Multi-class Classification

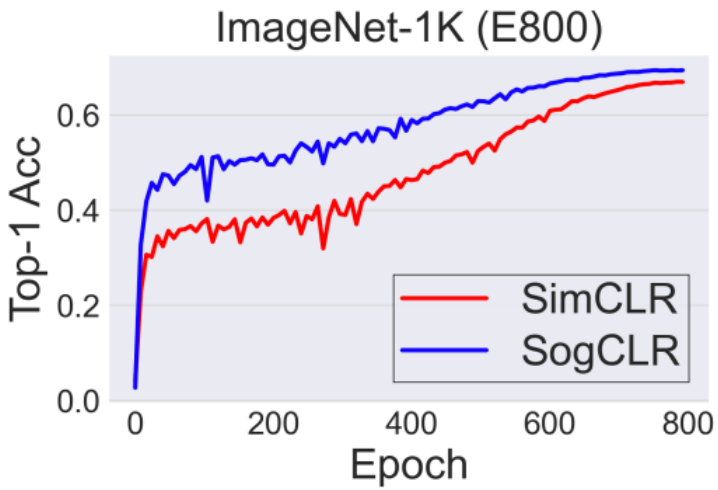


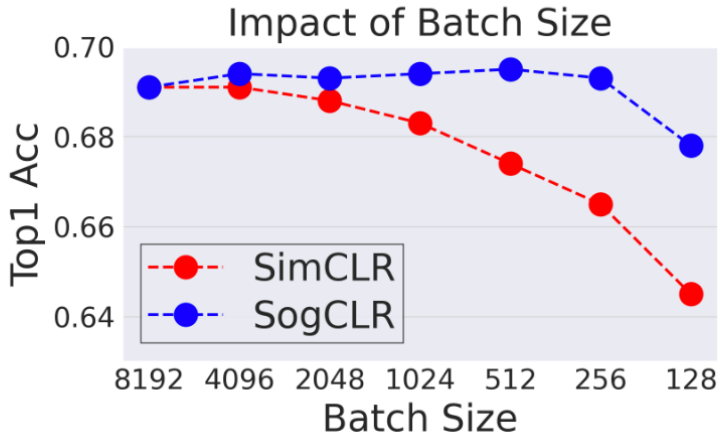
Glint360K dataset (An et al., 2021): a face recognition dataset consisting of n=17 million images of K=360 thousand individuals

Wei et al. (ICML'26)

SSL: SogCLR vs Google SimCLR

1.2M ImageNet Images

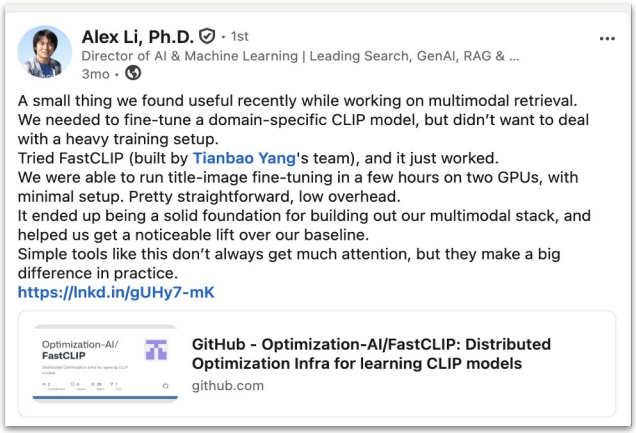




Yuan et al. (ICML'22)

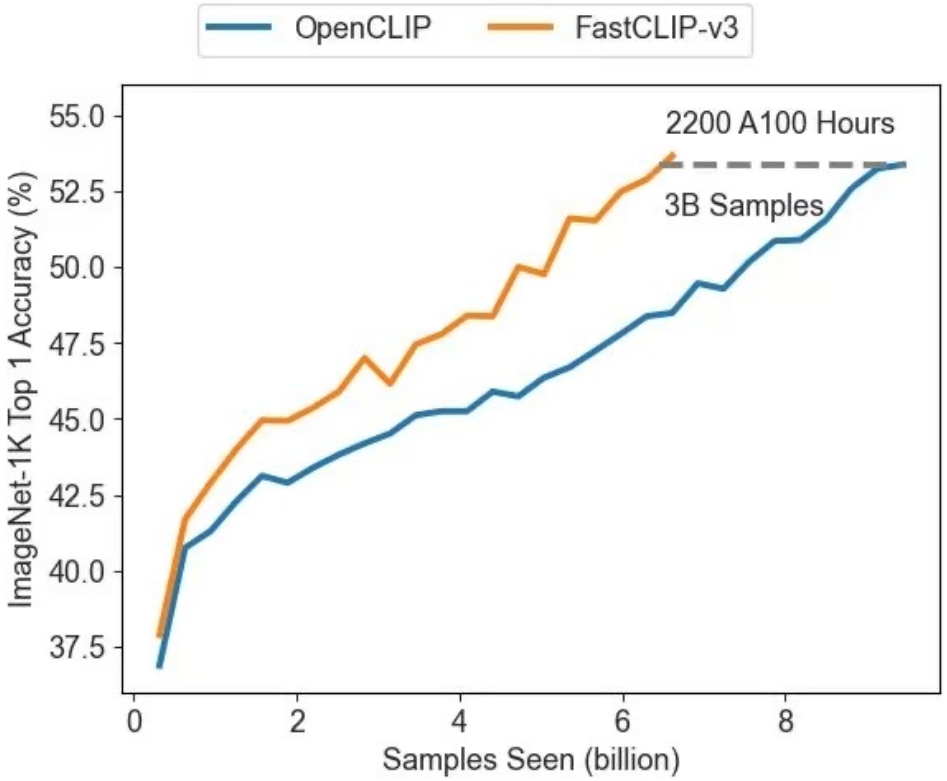
CLIP: FastCLIP vs OpenCLIP

9B samples seen
315M image-text pairs
8 H100s



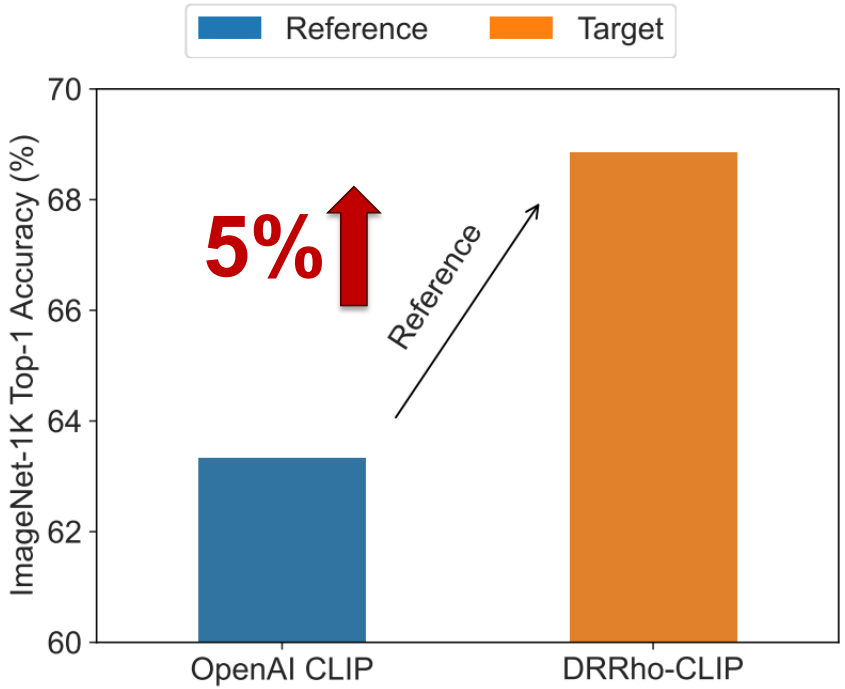
Adopted by Target

Wei et al. ’24



DRRHO-CLIP: reference guided

$$\ell_{ij}(\mathbf{w}) - \ell_{ij}(\mathbf{w}_r)$$



OpenAI: 400M data, 12.8B samples seen and 32768 batch size

2x less data size

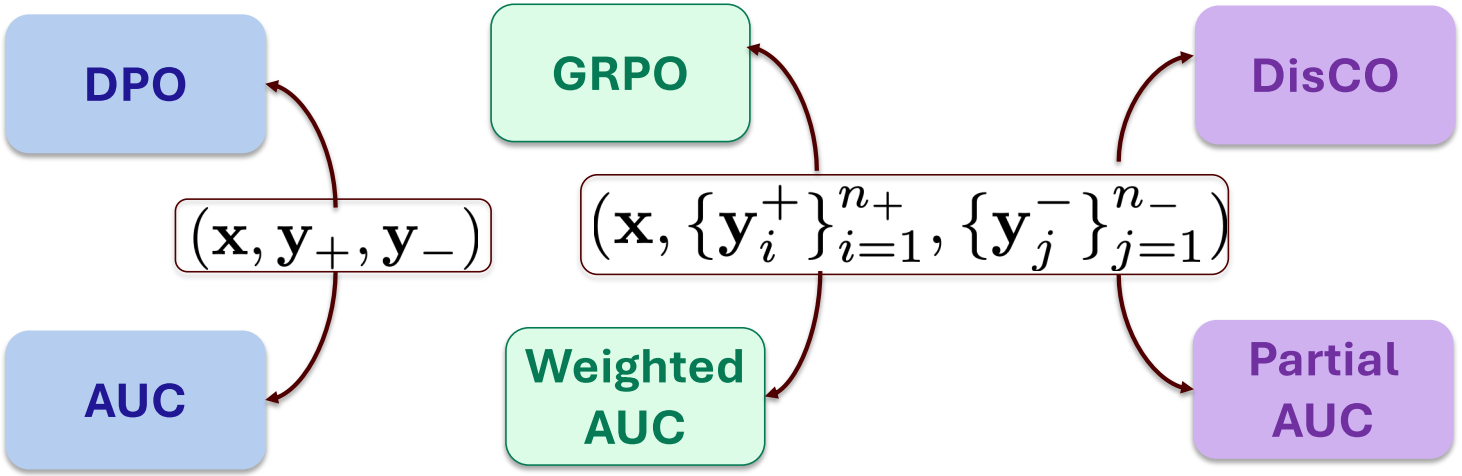
6x less batch size

17x less compute budget

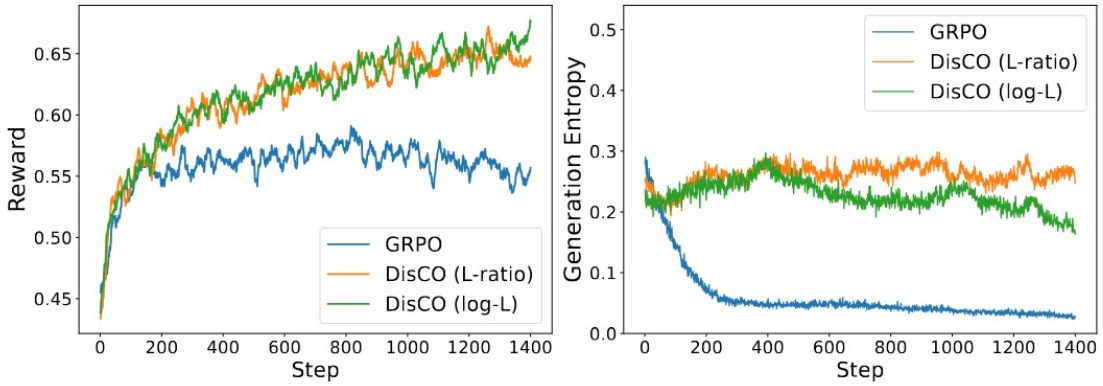
DRRHO-CLIP: 192M data, 1.28B samples seen and 5120 batch size

Wei et al. (ICML'25)

Post-training of LLMs



$$s(\mathbf{w}; \mathbf{x}, \mathbf{y}) = \log p_{\mathbf{w}}(\mathbf{y}|\mathbf{x})$$



Li et al. (NeurIPS'25)

libauc.org



LibAUC

A Deep Learning Library for X-Risk Optimization

An open-source library that translates theories to real-world applications

Latest News | Install

[2026-05] LibAUC 2.0 is released!

3+

Challenges winning solution (e.g., Stanford CheXpert, MIT AICures, OGB Graph Property Prediction).

4+

Collaborations and Deployments at multiple industrial units, e.g., Google, Uber, Tencent, etc.

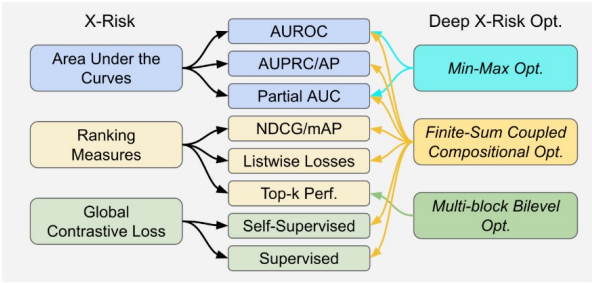
25+

Scientific publications on top-tier AI Conferences (e.g., ICML, NeurIPS, ICLR).

Why LibAUC?

LibAUC is a novel deep learning library to offer an easier way to directly optimize commonly used performance measures and losses with user-friendly APIs. LibAUC has broad applications in AI for tackling both classic and emerging challenges, such as *Classification of Imbalanced Data (CID)*, *Learning to Rank (LTR)*, and *Contrastive Learning of Representation (CLR)*.

LibAUC provides a unified framework to abstract the optimization of a family of risk functions called **X-Risk**, including surrogate losses for *AUROC*, *AUPRC/AP*, and *partial AUROC* that are suitable for CID, surrogate losses for *NDCG*, *top-K NDCG*, and *listwise losses* that are used in LTR, and *global contrastive losses* for CLR. For more details, please check our LibAUC paper.



133,000+

LibAUC Training Pipeline



1 **Compute Dynamic mini-batch loss**

$$\frac{1}{B_1} \sum_{\mathbf{x}_i \in \mathcal{B}_1^t} \nabla f_i(u_{i,t}) g(\mathbf{w}_t, \mathbf{x}_i, \mathcal{B}_i)$$

2 **Backpropagate gradients**

```
loss.backward()
```

3 **Update model parameters**

```
optimizer.step()
```

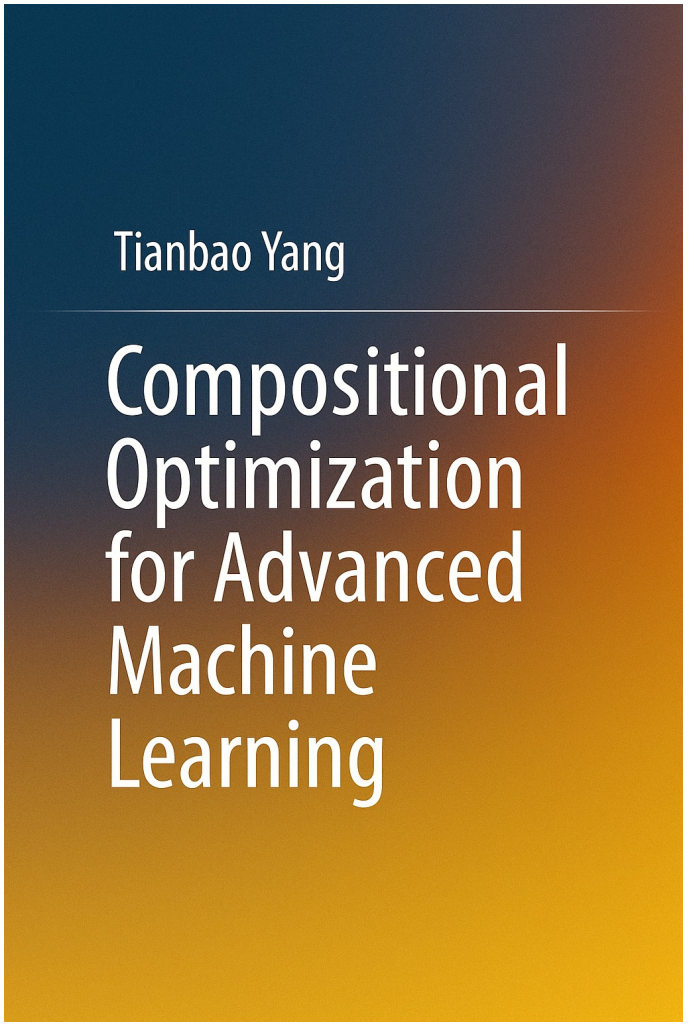
Take-home messages

- **Empirical X-risk Minimization**
  - A robust learning framework
  - A compositional optimization framework
- **LibAUC library**
  - Theory-guided, Broad applicability
- **DO NOT Abandon Contrastive methods**
  - Use EXM to formulate and solve them

“Let practice inspire theory, and let theory transform practice”

opt4ml.org




Tianbao Yang
Compositional Optimization for Advanced Machine Learning

Q&A